

SET - A

Q1 - C

Q1. $\frac{E(s)}{R(s)} = ?$

$$G = \frac{C(s)}{E(s)}$$

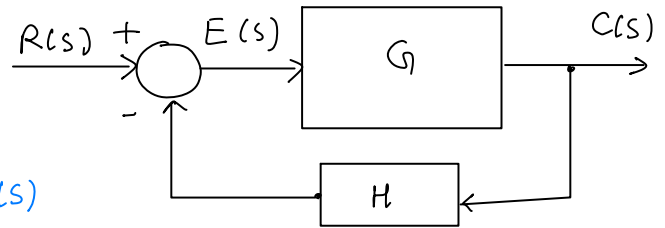
$$E(s) = R(s) - H \times C(s)$$

$$E(s) = R(s) - H \times G \times E(s)$$

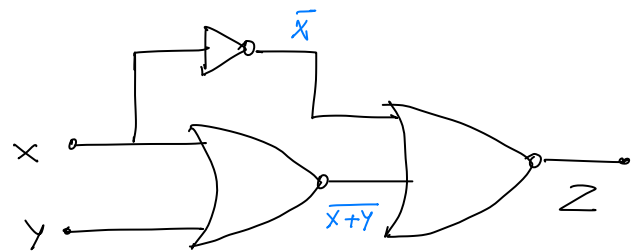
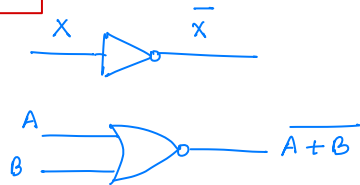
$$E(s) + H \times G \times E(s) = R(s)$$

$$E(s) \{ 1 + HG \} = R(s)$$

$$C) \quad \frac{E(s)}{R(s)} = \frac{1}{1 + GH}$$



Q2 - B



$$Z = \overline{\bar{X} + \overline{X+Y}}$$

$$= \overline{\bar{X} + \bar{X} \cdot \bar{Y}}$$

$$= (\overline{\bar{X}}) \cdot (\overline{\bar{X} \cdot \bar{Y}})$$

$$= X \cdot (\bar{X} + \bar{\bar{Y}}) = X(X + Y) = X \quad \text{[Absorption law]}$$

B) $Z = X$

$$Z = \overline{\bar{X} + \overline{X+Y}}$$

Q3.

$$XY + XZ' + YZ$$

Q3 - A

$$YZ + XZ' + XY(Z + Z') \quad \because Z + Z' = 1$$

$$YZ + XYZ + XZ' + XYZ'$$

$$= YZ(1 + X) + XZ'(1 + Y)$$

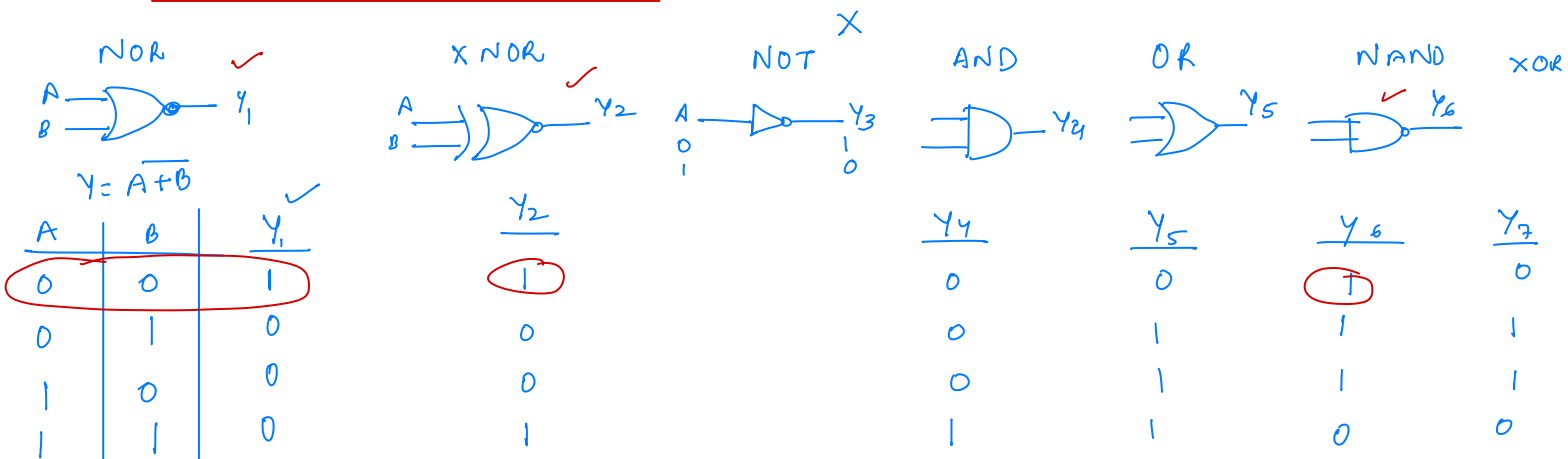
$$= YZ + XZ'$$

$$YZ + X\bar{Z}$$

$$\{ 1 + A = 1 \}$$

$$XY + YZ' + YZ = YZ + XZ'$$

Q4) D- NOR or X-NOR



Q5) B) Light Switch (1, 0)
on / off

Q6) C) Q and R P → False

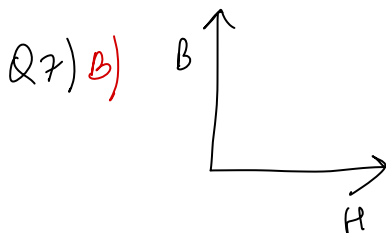
Analog more susceptible to noise: any small variation in the s/l can distort the info.

Q → True

Analog s/l's are continuous & can capture an ∞ range of values, resulting in higher resolution

R → True

S → False. Digital signals have limited resolution determined by the bits depth, while analog signals can represent a continuous range of values.



Q8) B)

Q9) 1 1 1 1 1 1 0 0 0
 $\underbrace{\hspace{1cm}}$
 MSB = 1 ∴ no. is -ve

$$\begin{array}{r} \text{invert} \\ 00000111 \\ + \quad \quad \quad 1 \\ \hline 00001000 = 8 \text{ in decimal} \end{array}$$

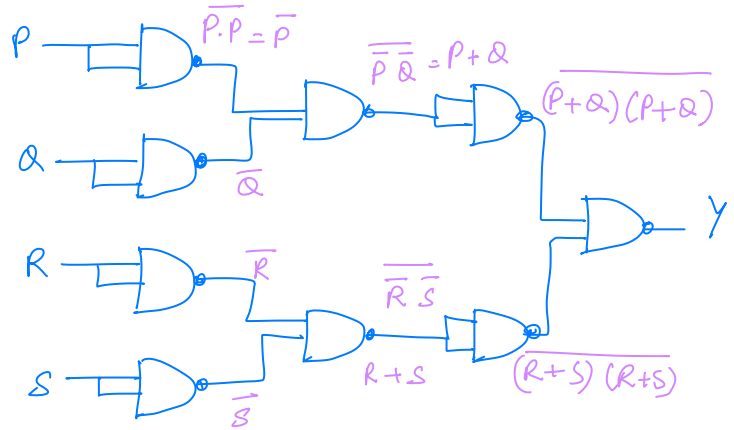
Q9: -8

∴ -8

Q10)

$$P + Q + R + S$$

$$\begin{aligned}
 Y &= \overline{(P+Q)(P+Q) \cdot (R+S)(R+S)} \\
 &= \overline{(P+Q)} \cdot \overline{(R+S)} \\
 &= P+Q + R+S
 \end{aligned}$$



Q11)

$$(101010)_2 = 42$$

Q12)

$$3.98 \times 10^6 \text{ A/Wb}$$

$$A = 50 \times 10^{-4} \text{ m}^2$$

$$R = \frac{l}{\mu_0 \mu_r A} = \frac{2}{80 \times 4\pi \times 10^{-7} \times 0.005}$$

Q13)

$$6.7 \text{ A}$$

$$A = 500 \text{ mm}^2 = 500 \times 10^{-6} \text{ m}^2, \quad l = 400 \text{ mm} = 400 \times 10^{-3} \text{ m}$$

$$\phi = 800 \times 10^{-6} \text{ Wb}, \quad N = 200, \quad \mu_r = 380$$

$$B = \frac{\phi}{A}, \quad H = \frac{B}{\mu_0 \mu_r}, \quad F = NI = Hl$$

$$I = \frac{Hl}{N} = \frac{3353.9 \times 400 \times 10^{-3}}{200}$$

$$\begin{aligned}
 &\downarrow \\
 &= \frac{800 \times 10^{-6}}{500 \times 10^{-6}} = \frac{1.6 \text{ T}}{(4\pi \times 10^{-7})(380)} \\
 &= 1.6 \text{ T} = 3353.9 \text{ A/m}
 \end{aligned}$$

$$I \approx \frac{13.4 \text{ A}}{2} = 6.7 \text{ A}$$

Q14.

$$20 \text{ V}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$V_s = V_p \times \frac{N_s}{N_p} = 200 \text{ V} \times \frac{20}{200} = 20 \text{ V}$$

Q15.

$$4.8$$

$$S = V_1 I_1$$

$$I_1 = \frac{S}{V_1} = \frac{12 \times 10^3 \text{ VA}}{2500 \text{ V}} = 4.8 \text{ A}$$

$$\left. \begin{array}{l} \\ \end{array} \right\} V_1 = 2.5 \text{ k} = 2500$$